REGRESSION

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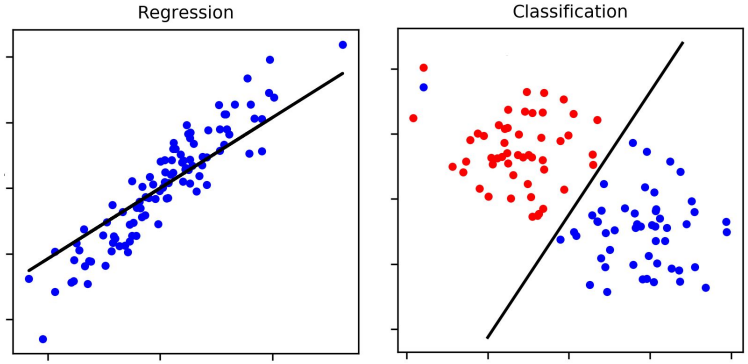
Regression

- A supervised ML task in which the model predict the continuous or numerical outcome
- Regression task can be further divided into:
  - “Normal” regression
  - Count regression
  - Time-series
  - Censored regression

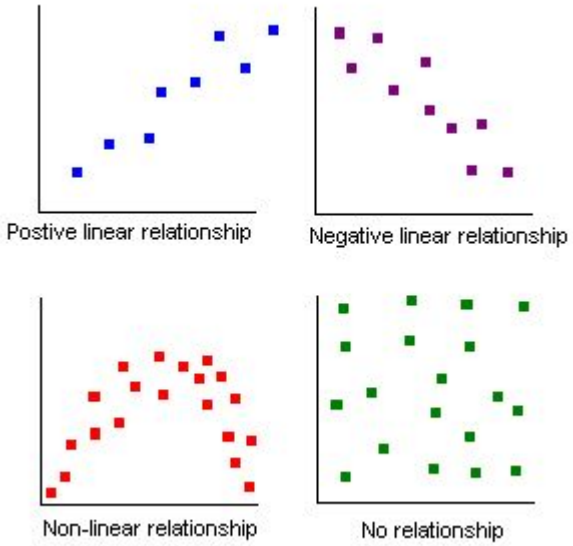


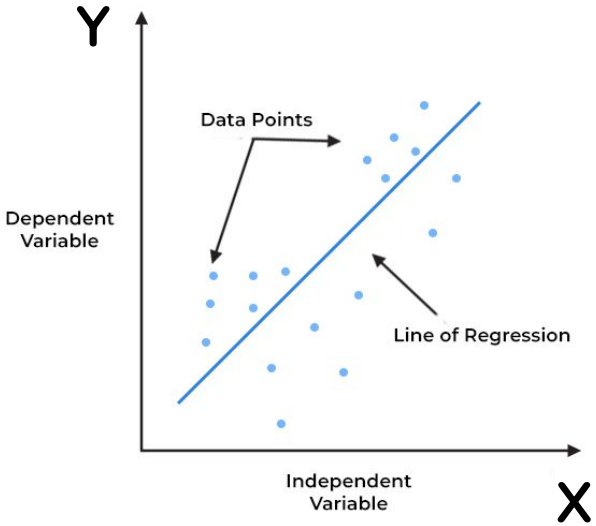
Regression algorithms

- Just to list a few:
  - LinearRegression
  - Decision Trees
  - Random Forest
  - Support Vector Machines (SVM)
  - Partial Least Squares
  - Artificial Neural Networks (ANN)
- Full list of algorithms in parsnip package

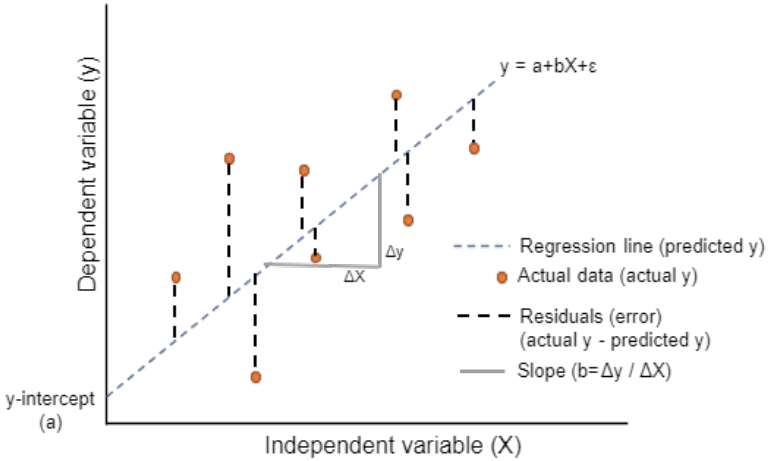
Linear regression

- Aims: to describe the **linear relationship** between a numerical outcome variable and independent variables/predictors



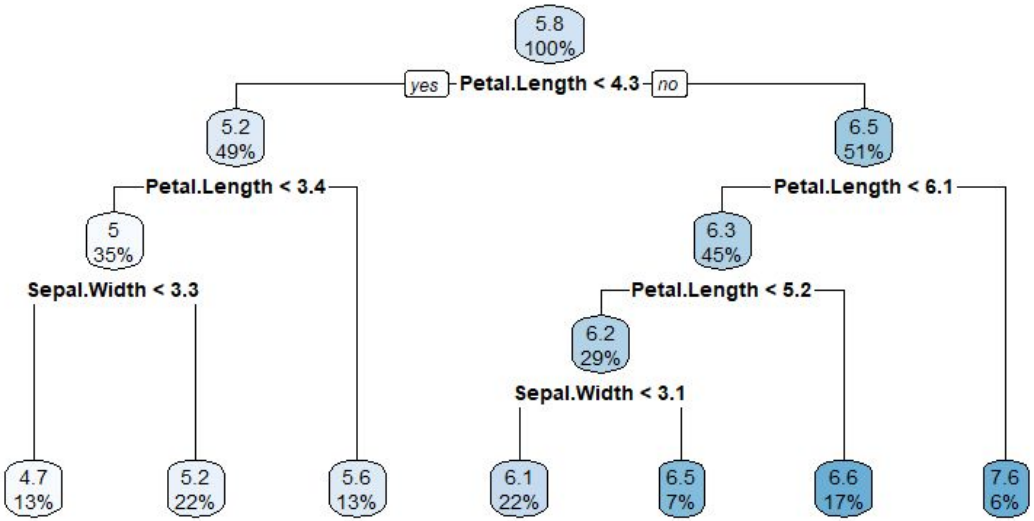


- Concept/idea
  - To find the **best fit line** to describe the overall data
    - The line that minimise the difference between the observed and the predicted values
    - $Y = \beta_0 + \beta_1 X_1 + \varepsilon$ (equivalent to $Y = mX + b + \varepsilon$)

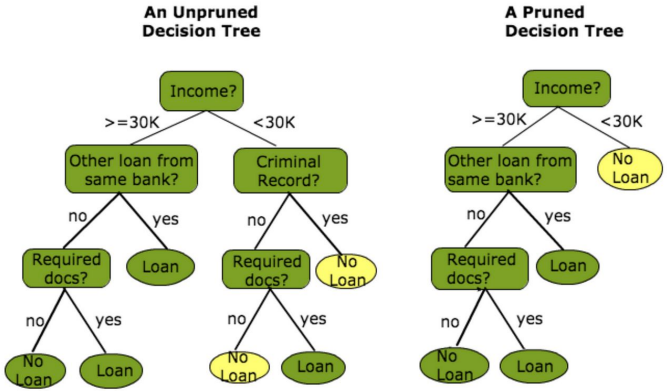


Decision tree

- Can be used for regression and classification - classification and regression trees (CART) models
- The split is determined by the one that gives the lowest error (to be exact the variance) of the target variable within a subset



- Pros:
  - Easy to understand
  - Fast computation
  - Able to handle missing data and outliers
- Cons:
  - Tree overfitting - can be overcome with a pruning or CV methods



Performance metrics

RMSE

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

- RMSE - root mean squared error
- It reflects the error in predicting the outcome and more sensitive to large error
- The same unit as the data

MSE

$$MSE = \frac{\sum(y_i - \hat{y}_i)^2}{n}$$

- MSE - mean squared error
- Not the same unit as our data
- Heavily penalise large errors due to squarring
- Less interpretable as compared RMSE and MAE

MAE

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\widehat{y_i} - y_i|$$

- MAE - mean absolute error
- It reflects the error in predicting the outcome and treat the error equally (as opposed to RMSE)
- It is in the same unit as the data

Huber loss

$$Huber = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{2}(y_i - \hat{y}_i)^2 \qquad |y_i - \hat{y}_i| \leq \delta$$

$$Huber = \frac{1}{n}\sum_{i=1}^{n}\delta\left(|y_i - \hat{y}_i| - \frac{1}{2}\delta\right) \qquad |y_i - \hat{y}_i| > \delta$$

- It combine MSE and MAE best properties:
  - For small errors (|error| ≤ δ): use MSE
  - For large errors (|error| > δ): use MAE
- Robust loss function - less sensitive to outliers and extreme values

- Useful when data contains a lot of outliers or noisy data
- There is pseudo huber loss in tidymodels:
    - Basically, pseudo huber loss approximate the huber loss but has a stable and more efficient computation

R-square

$$R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

- How much the predictors are able to explain the proportion of the variance in the outcome
- It reflect the goodness of fit of the model
- It ranges between 0 (poor fit) to 1 (good fit)

CCC

- CCC - concordance correlation coefficient
- It evaluates both:
  - Consistency/precision: how well the predicted values agree with the true values
  - Accuracy: how close the predicted values are to the true values
- For example, we have true values of [1, 2, 3] and two sets of predictions:
  - Set A: [1.1, 2.1, 3.1]
    - This set is consistent (all predictions are off by 0.1) but not accurate (they are all slightly off the true values).
  - Set B: [1, 2, 4]
    - This set is accurate for some values but inconsistent (the third prediction is quite off).

- CCC - the higher the better
- Interpret:
  - CCC close to 1: good model (consistent and accurate)
  - CCC close 0: poor model
  - Negative CCC values imply systematic disagreement

Suggested readings/references

- Kuhn, M., & Silge, J. (2022). Tidy Modeling with R: A Framework for Modeling in the Tidyverse. O'Reilly Media.
- Burger, S. V. (2018). Introduction to machine learning with R: Rigorous mathematical analysis (First edition). O'Reilly Media.

Any question?

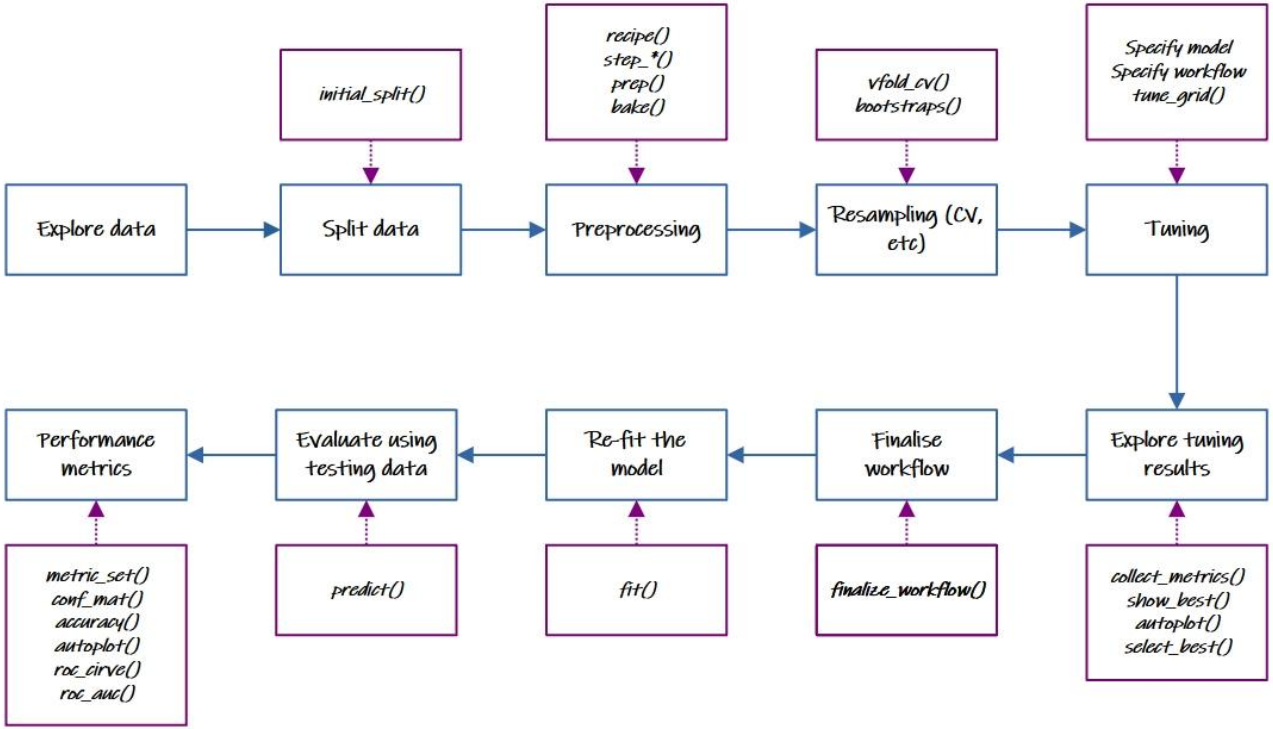
Hands-on in R

Tidymodels

- Tidymodels - a collection of packages for modeling and machine learning using a tidyverse principles
  - Tidyverse - an opinionated collection of R packages designed for data science
- Some of the core packages:
  - rsample - data splitting and resampling
  - parsnip - unify various ML models/algorithms from different packages
  - recipes - data pre-processing tools for feature engineering
  - tune - hyperparameter tuning
  - yardstick - assess the performance of ML models

- Other individual and small packages are available as well like nnet, kernlab, etc
- Alternatives to tidymodels:
  - caret - R only
  - mlr3 - R only
  - h2o - available in python and R

Basic workflow in tidymodels



Data - bike-share.csv

- This dataset contains the **daily count of rental bikes** between the years 2011 and 2012 in the Capital bike share system with the corresponding weather and seasonal information
- More info - https://www.kaggle.com/datasets/lakshmi25npathi/bike-sharing-dataset?select=day.csv

- Variables:
  - instant: identifier
  - dteday: date
  - season: 1-spring, 2-summer, 3-fall, 4-winter
  - yr: year 0 = 2011, and year 1 = 2012
  - mnth: month
  - holiday: binary - public holiday or not
  - weekday: 0-Sunday … 6-Saturday
  - workingday: binary - working day or not (regardless of weekend or holiday)
  - weathersit: 1-clear, 2-mist/cloud, 3-light rain/snow, 4-heavy rain/hail/snow/fog
  - temp: temperature in celsius (normalized)
  - atemp: apparent (“feels-like”) temperature in celsius (normalized)
  - hum: humidity level (normalized)
  - windspeed: windspeed (normalized)
  - rentals: number of bicycle rentals recorded (outcome)

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